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Modified likelihood ratio tests for unit gamma regressions

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ABSTRACT

Regression analyses are commonly performed with doubly limited continuous dependent variables; for instance, when modeling the behavior of rates, proportions and income concentration indices. Several models are available in the literature for use with such variables, one of them being the unit gamma regression model. In all such models, parameter estimation is typically performed using the maximum likelihood method and testing inferences on the model's parameters are usually based on the likelihood ratio test. Such a test can, however, deliver quite imprecise inferences when the sample size is small. In this paper, we propose two modified likelihood ratio test statistics for use with the unit gamma regressions that deliver much more accurate inferences when the number of data points is small. Numerical (i.e. simulation) evidence is presented for both fixed dispersion and varying dispersion models, and also for tests that involve nonnested models. We also present and discuss two empirical applications.

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1. Introduction

Regression analysis is commonly used to explain the behavior of doubly limited continuous dependent variables (DBCDVs) that assume values in (a, b) , where a and b are known and $-\infty < a < b < \infty$. It is well known that the linear regression model is not appropriate for modeling such variables since it may yield fitted values that are smaller than a or larger than b . In addition, it fails to account for distributional asymmetry and heteroskedasticity. There is then need for regression models that are tailored to DBCDVs. Most frequently, the interest lies in modeling variables that assume values in the standard unit interval, i.e. $a = 0$ and $b = 1$. This is commonly done using the beta regression model introduced by Ferrari and Cribari-Neto [15] and later generalized to include varying dispersion by Simas *et al.* [31]. Many empirical studies in different fields have been based on the beta regression model, such as [6,21,22,25,34,36].

The beta regression model is the most well known and the most widely used model with DBCDVs. Alternative models have, nonetheless, been introduced in the literature since it is useful for practitioners to have more than a single model at disposal. For example, Lemonte and Bazán [24] proposed the Johnson S_B regression model. Unlike the beta

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regression model, it allows users to model median effects. Another alternative regression model was proposed by Mousa *et al.* [27]: the unit gamma regression model. Like the beta model, here one models mean effects. The underlying assumption is that the variable of interest follows the unit gamma law introduced by Grassia [18]. Parameter estimation in such models is typically carried out by maximum likelihood and testing inferences are generally performed using the likelihood ratio test.

The unit gamma law can be a useful alternative to the beta law in some applications. Consider, e.g. the application outlined by Grassia [18] that involves using the inoculation approach to estimate bacteria or virus density in dilution assays with host variability to infection. The use of the beta density results in an expression for the probability of no infection (which the author denotes by Q) that cannot be easily fitted to data. The unit gamma law is obtained in that context by making use of simple arguments and yields a simple expression for desired probability. This example shows that it is useful for practitioners to have more than one model at their disposal. The beta distribution and the beta regression model are certainly the most commonly used models for DBCDVs. There are always situations, nonetheless, in which an alternative model shows relative merits, as in the aforementioned application. In this paper we pursue one of such alternative models, namely: the unit gamma regression model.

The likelihood ratio test is the most commonly used test in regression analysis. Such a test, however, can be considerably size-distorted in small samples. A well known correction to the likelihood test statistic is the Bartlett correction [23]. However, it requires the derivation of a large number of log-likelihood cumulants and involve long and tedious algebra. In this paper we shall make use of a simpler approach: we shall use a correction to the likelihood ratio test statistic that only requires second-order derivatives of the log-likelihood function.

Several regression analyses with DBCDVs are based on samples of small to moderate sizes. For instance, Espinheira *et al.* [13] present a beta regression analysis that includes modeling and response prediction based on only 42 observations. Their interest lies in modeling a simultaneity factor which is used to determine the distribution of natural gas for home usage (in stoves, ovens, water heaters, etc.) in São Paulo, Brazil. Additionally, Souza and Cribari-Neto [35] use the beta regression model to measure the impact of average intelligence on the prevalence of religious disbelievers in the United States. Since they use state level data, the analysis is based on only 50 observations. Finally, in Section 5 we shall measure the impact of dyslexia on reading accuracy based on only 44 data points. As we shall see, there is some evidence that the unit gamma regression model fits such data better than the commonly used beta regression model. Such an empirical application indicates that it is important to have at disposal testing inference strategies that are accurate in small samples. We also model an international cross section dataset in which the variable of interest are the proportions of atheists in over one hundred countries. The results favor the unit gamma model more clearly than those of the previous application.

In this paper we shall focus on the unit gamma regression model. We shall consider mean effects modeling and also joint mean and precision effects modeling. Our interest is on hypothesis testing inferences performed with samples of small sizes. In particular, our chief goal is to derive two modified likelihood ratio test statistics that can be used to perform reliable testing inferences in unit gamma regressions when the sample size is small. The Monte Carlo evidence that we present shows that testing inferences based on the two

modified test statistics can be considerably more accurate than that based on the standard likelihood test statistic in small samples. The latter can be considerably oversized. Hence, by using the standard likelihood ratio test in unit gamma regressions with small samples practitioners may erroneously and frequently conclude that some regressors should be included in the model when in fact they are not relevant for explaining the behavior of the variable of interest. Tests with improved finite sample behavior are obtained for use with beta regressions. To the best of our knowledge, however, no similar results have been obtained for the unit gamma regression model.

The paper is structured as follows. Section 2 presents the beta and unit gamma regression models. In Section 3, we derive two test statistics that can be used to perform testing inferences in unit gamma regressions. They are expected to deliver testing inferences that are more accurate than those achieved through the likelihood ratio test when the sample size is small. Section 4 contains Monte Carlo simulation results on the finite sample performances of the two tests we propose and also on that of the standard likelihood ratio test. Two empirical applications are presented and discussed in Section 5. Finally, Section 6 contains some concluding remarks.

2. The unit gamma and beta regression models

The beta distribution is commonly used to model continuous data that are restricted to the standard unit interval. Its density can be left- or right-skewed, symmetric, J-shaped and inverted J-shaped. Ferrari and Cribari-Neto [15] proposed a regression model based on the assumption that the response (y) is beta-distributed. The authors changed the distribution parameterization so that it becomes indexed by location (μ) and precision (ϕ) parameters, i.e. $y \sim \text{Beta}(\mu, \phi)$. The reparameterized beta density is

$$b(y; \mu, \phi) = \frac{\Gamma(\phi)}{\Gamma(\mu\phi)\Gamma(\phi(1-\mu))} y^{\mu\phi-1} (1-y)^{\phi(1-\mu)-1}, \quad 0 < y < 1,$$

$0 < \mu < 1, \phi > 0$, where $\Gamma(\cdot)$ is the gamma function. Here, $\mathbb{E}(y) = \mu$ and $\text{Var}(y) = \mu(1-\mu)/(\phi+1)$.

Let $y_1, \dots, y_n$ be independent random variables, each y_i following the beta law, i.e. $y_i \sim \text{Beta}(\mu_i, \phi_i)$, $i = 1, \dots, n$. The location (mean) and precision submodels are

$$g_1(\mu_i) = \eta_i = \sum_{j=1}^p \beta_j x_{ij} \quad \text{and} \quad g_2(\phi_i) = \zeta_i = \sum_{j=1}^q \delta_j h_{ij},$$

respectively, where $\beta = (\beta_1, \dots, \beta_p)^\top$ and $\delta = (\delta_1, \dots, \delta_q)^\top$ are vectors of unknown regression parameters $\beta \in \mathbb{R}^p$ and $\delta \in \mathbb{R}^q$, $x_{i1} \equiv h_{i1} \equiv 1$, and $x_{i2}, \dots, x_{ip}$ and $h_{i2}, \dots, h_{iq}$ are observations on p and q covariates ($p+q < n$). Finally, $g_1(\cdot)$ and $g_2(\cdot)$ are strictly monotonic and twice differentiable functions that map $(0, 1) \rightarrow \mathbb{R}$ and $(0, \infty) \rightarrow \mathbb{R}$, respectively. Possible choices of $g_1(\mu_i)$ are: logit $g_1(\mu_i) = \log(\mu_i/(1-\mu_i))$, probit $g_1(\mu_i) = \Phi^{-1}(\mu_i)$, where $\Phi(\cdot)$ is the standard normal cumulative distribution function, log-log $g_1(\mu_i) = -\log(-\log(\mu_i))$, complementary log-log $g_1(\mu_i) = \log(-\log(1-\mu_i))$, and Cauchy $g_1(\mu_i) = \tan(\pi(\mu_i - 0.5))$. Possible choices of $g_2(\phi_i)$ are: identity $g_2(\phi_i) = \phi_i$, log $g_2(\phi_i) = \log(\phi_i)$ and square root $g_2(\phi_i) = \sqrt{\phi_i}$. In the fixed dispersion beta regression model, $g_2(\phi_i) = g_2(\phi) = \delta_1$.

An alternative regression model was introduced by Mousa *et al.* [27]. It is based on the unit gamma distribution proposed by Grassia [18]. Like the beta regression model, it is useful for modeling DBCDVs. The unit gamma distribution is indexed by two parameters: $\alpha, \phi > 0$. We shall denote it by $ug(\alpha, \phi)$. The unit gamma density can be written as

$$ug(y; \alpha, \phi) = \frac{\alpha^\phi}{\Gamma(\phi)} y^{\alpha-1} \left[\log \left(\frac{1}{y} \right) \right]^{\phi-1}, \quad 0 < y < 1,$$

$\alpha > 0, \phi > 0$. Here, $\mathbb{E}(y) = [\alpha/(\alpha + 1)]^\phi$ and $\text{Var}(y) = [\alpha/(\alpha + 2)]^\phi - [\alpha/(\alpha + 1)]^{2\phi}$. The density was reparameterized by setting $\alpha = [\mu^{1/\phi}/(1 - \mu^{1/\phi})]$ so that $\mathbb{E}(y) = \mu$ and $\text{Var}(y) = \mu\{[1/(2 - \mu^{1/\phi})^\phi] - \mu\}$. Notice that the variance is a function of the mean. The new density function is

$$ug(y; \mu, \phi) = \frac{\left(\frac{\mu^{1/\phi}}{1 - \mu^{1/\phi}}\right)^\phi}{\Gamma(\phi)} y^{\frac{\mu^{1/\phi}}{1 - \mu^{1/\phi}} - 1} \left[\log \left(\frac{1}{y} \right) \right]^{\phi-1}, \quad 0 < y < 1,$$

$0 < \mu < 1, \phi > 0$. We shall denote the reparameterized unit gamma distribution by $ug(\mu, \phi)$. Notice that ϕ can be interpreted as a precision parameter.

Let $y_1, \dots, y_n$ be independent random variables, where each $y_i \sim ug(\mu_i, \phi_i)$, $i = 1, \dots, n$, with mean μ_i and precision ϕ_i . In the unit gamma regression model proposed by Mousa *et al.* [27], the i th mean response and the i th precision can be written as

$$g_1(\mu_i) = \eta_i = \sum_{j=1}^p \beta_j x_{ij} \quad \text{and} \quad g_2(\phi_i) = \zeta_i = \sum_{j=1}^q \delta_j h_{ij}, \quad (1)$$

respectively. Since $\text{Var}(y_i) = \mu_i\{[1/(2 - \mu_i^{1/\phi_i})^\phi] - \mu_i\}$, where $\mu_i = g_1^{-1}(\eta_i)$, the regression model is heteroskedastic. The fixed dispersion unit gamma regression model is obtained by setting $g_2(\phi_i) = g_2(\phi) = \delta_0$.

Consider the unit gamma regression model defined in (1) and write the model parameter vector as $\theta = (\beta^\top, \delta^\top)^\top$. The varying dispersion unit gamma regression model log-likelihood function is

$$\ell(\theta) \equiv \ell(\beta, \delta) = \sum_{i=1}^n \ell_i(\mu_i, \phi_i), \quad (2)$$

where $\ell_i(\mu_i, \phi_i) = \phi_i \log(d_i) - \log \Gamma(\phi_i) + (d_i - 1)y_i^* + (\phi_i - 1)y_i^\dagger$. Here,

$$y_i^* = \log(y_i), \quad y_i^\dagger = \log(-\log(y_i)) \quad \text{and} \quad d_i = \mu_i^{1/\phi_i}/(1 - \mu_i^{1/\phi_i}). \quad (3)$$

It is known that $-y^*$ is gamma-distributed with parameters ϕ and d . It then follows that $\mathbb{E}(y^*) = \mathbb{E}(-z) = -\phi/d$. Additionally, $y^\dagger = \log(-\log(y)) = \log(z)$ is distributed as log-gamma with parameters ϕ and d . It can then be shown that

$$\begin{aligned} \mu_i^* = \mathbb{E}(y_i^*) &= -\frac{\phi_i}{d_i}, & \mu_i^\dagger = \mathbb{E}(y_i^\dagger) &= \psi(\phi_i) - \log(d_i), & v_i^* = \text{Var}(y_i^*) &= \frac{\phi_i}{d_i^2}, \\ v_i^\dagger = \text{Var}(y_i^\dagger) &= \psi'(\phi_i), & c_i^{*\dagger} &= -\frac{1}{d_i}, \end{aligned}$$

where $\psi(\cdot)$ and $\psi'(\cdot)$ are the digamma and trigamma functions, respectively.

The log-likelihood function $\ell(\theta)$ can be written in matrix form as

$$\ell(\theta) = \{(y^* - \mu^*)^\top (D - \mathcal{I}) + (y^\dagger - \mu^\dagger)^\top (\Phi - \mathcal{I}) + b^\top\} \iota, \quad (4)$$

where $y^* = (y_1^*, \dots, y_n^*)^\top$, $\mu^* = (\mu_1^*, \dots, \mu_n^*)^\top$, $y^\dagger = (y_1^\dagger, \dots, y_n^\dagger)^\top$, $\mu^\dagger = (\mu_1^\dagger, \dots, \mu_n^\dagger)^\top$, $D = \text{diag}(d_1, \dots, d_n)$, $\Phi = \text{diag}(\phi_1, \dots, \phi_n)$, $\mathcal{I}$ is the $n \times n$ identity matrix, ι is the n -dimensional column vector of ones and $b = (b_1, \dots, b_n)^\top$ with $b_i = \phi_i \log(d_i) - \log \Gamma(\phi_i) + \mu_i^*(d_i - 1) + \mu_i^\dagger(\phi_i - 1)$, $i = 1, \dots, n$, with d_i defined in (3).

It follows from (1) that

$$\frac{\partial \ell(\beta, \phi)}{\partial \beta_j} = \sum_{i=1}^n \left\{ \frac{d_i}{\mu_i \phi_i} (1 + d_i) (y_i^* - \mu_i^*) \frac{1}{g'_1(\mu_i)} x_{ij} \right\}, \quad (5)$$

$j = 1, \dots, p$. In similar fashion,

$$\frac{\partial \ell(\beta, \phi)}{\partial \delta_j} = \sum_{i=1}^n \left\{ \left[\left(\frac{-d_i \log(\mu_i)(1 + d_i)}{\phi_i^2} \right) (y_i^* - \mu_i^*) + (y_i^\dagger - \mu_i^\dagger) \right] \frac{1}{g'_2(\phi_i)} h_{ij} \right\}, \quad (6)$$

$j = 1, \dots, p$.

In what follows we shall write (5) and (6) in matrix form. Some matrices we use in the formulas that follow are defined in the Appendix. The score function, obtained by differentiating the log-likelihood function with respect to the unknown parameters, is given by $U = (U_\beta(\beta, \delta)^\top, U_\delta(\beta, \delta)^\top)^\top$, where

$$U_\beta(\beta, \delta) = X^\top T_1 \Phi^{-1} \mathcal{M}^{-1} D (\mathcal{I} + D) (y^* - \mu^*) \quad (7)$$

and

$$U_\delta(\beta, \delta) = H^\top T_2 [P(y^* - \mu^*) + (y^\dagger - \mu^\dagger)], \quad (8)$$

where $T_1 = \text{diag}(1/g'_1(\mu_1), \dots, 1/g'_1(\mu_n))$, $T_2 = \text{diag}(1/g'_2(\phi_1), \dots, 1/g'_2(\phi_n))$, $\mathcal{M} = \text{diag}(\mu_1, \dots, \mu_n)$, $D = \text{diag}(d_1, \dots, d_n)$, $P = \text{diag}(-d_1(1 + d_1) \log(\mu_1)/\phi_1^2, \dots, -d_n(1 + d_n) \log(\mu_n)/\phi_n^2)$, X is an $n \times p$ matrix whose i th row is $x_i^\top$ and H is $n \times q$ matrix whose i th row is $h_i^\top$. The maximum likelihood estimators of β and δ solve $U_\beta(\beta, \delta) = U_\delta(\beta, \delta) = 0$. Maximum likelihood estimates are typically obtained by numerically maximizing the log-likelihood function using a Newton or quasi-Newton algorithm such as Broyden–Fletcher–Goldfarb–Shanno (BFGS) algorithm; see Nocedal and Wright [29].

The Hessian matrix, i.e. the matrix of second derivatives of the log-likelihood function with respect to the model's parameters, is

$$J = \begin{bmatrix} J_{\beta\beta} & J_{\beta\delta} \\ J_{\delta\beta} & J_{\delta\delta} \end{bmatrix},$$

where

$$J_{\beta\beta} = \frac{\partial^2 \ell(\beta, \delta)}{\partial \beta \partial \beta^\top} = X^\top [(K_1(Y^* - M^*) - K_2) T_1 - S_1 T_1^2 K_3 (Y^* - M^*)] T_1 X,$$

$$J_{\beta\delta} = \frac{\partial^2 \ell(\beta, \delta)}{\partial \beta \partial \delta^\top} = J_{\delta\beta}^\top = X^\top [Z_1(Y^* - M^*) + Z_2] T_1 T_2 H,$$

and

$$J_{\delta\delta} = \frac{\partial^2 \ell(\beta, \delta)}{\partial \delta \partial \delta^\top} = H^\top [T_2(L_1(Y^* - M^*) + L_2) - (L_3(Y^* - M^*) + (Y^\dagger - M^\dagger))S_2 T_2^\top] T_2 H.$$

Here, $Y^* = \text{diag}(y_1^*, \dots, y_n^*)$, $Y^\dagger = \text{diag}(y_1^\dagger, \dots, y_n^\dagger)$, $M^* = \text{diag}(\mu_1^*, \dots, \mu_n^*)$, $M^\dagger = \text{diag}(\mu_1^\dagger, \dots, \mu_n^\dagger)$, $S_1 = \text{diag}(g_1''(\mu_1), \dots, g_1''(\mu_n))$ and $S_2 = \text{diag}(g_2''(\phi_1), \dots, g_2''(\phi_n))$.

Fisher's information matrix is thus given by

$$I = I(\beta, \delta) = \begin{bmatrix} I_{\beta\beta} & I_{\beta\delta} \\ I_{\delta\beta} & I_{\delta\delta} \end{bmatrix},$$

where $I_{\beta\beta} = X^\top K_2 T_1^2 X$, $I_{\beta\delta} = I_{\delta\beta}^\top = -X^\top Z_2 T_1 T_2 H$ and $I_{\delta\delta} = -H^\top L_2 T_2^\top H$.

3. Modified likelihood ratio testing inference in unit gamma regressions

Consider the unit gamma regression model presented in (1) and also the corresponding log-likelihood function given in Equation (2), where $\theta = (\beta^\top, \delta^\top)^\top$ is the model k -dimensional parameter vector, β being a p -vector and δ being a q -vector such that $p + q = k$. In what follows, $\kappa = (\kappa_1, \dots, \kappa_l)^\top$ represents the parameter of interest and $\psi = (\psi_1, \dots, \psi_s)^\top$ is the nuisance parameter. (Note that $l + s = p + q$). Our interest thus lies in testing l restrictions. In particular, we wish to test $\mathcal{H}_0 : \kappa = \kappa^0$ versus $\mathcal{H}_0 : \kappa \neq \kappa^0$, where κ^0 is a fixed l -vector. The likelihood ratio test statistic can be written as $w = 2[\ell(\hat{\kappa}, \hat{\psi}) - \ell(\kappa^0, \tilde{\psi})]$, where $(\kappa^{0\top}, \tilde{\psi}^\top)$ and $(\hat{\kappa}^\top, \hat{\psi}^\top)$ are, respectively, the restricted and unrestricted maximum likelihood estimators of $(\kappa^\top, \psi^\top)$. Under the null hypothesis, w is asymptotically distributed as χ_l^2 . The null hypothesis is rejected at the α significance level ($0 < \alpha < 1$) if $w > \chi_{1-\alpha, l}^2$, where $\chi_{1-\alpha, l}^2$ is the $1 - \alpha$ upper χ_l^2 quantile.

When n is small, the approximation used in the likelihood ratio test may not be accurate, and as a result size distortions may take place. It is thus desirable to apply a finite sample correction to the test statistic when the sample size is small. A useful small sample correction was obtained by Skovgaard [33]. We shall use it as basis to obtain adjusted likelihood ratio test statistics whose distributions under $\mathcal{H}_0$ are well approximated by the χ^2 reference distribution even in small samples in the context of unit gamma regression modeling.

When the parameter of interest is scalar, the null hypothesis can be tested using the signed likelihood ratio test statistic:

$$w_s = \text{sign}(\hat{\kappa} - \kappa)[2\{\ell(\hat{\kappa}, \hat{\psi}) - \ell(\kappa, \tilde{\psi})\}]^{1/2},$$

where κ is the scalar parameter of interest and ψ is the vector of nuisance parameters. Under the null hypothesis, w_s is standard normally distributed with error of order $n^{-1/2}$. Such an approximation may be, nonetheless, inaccurate in small samples.

Barndorff-Nielsen [3,4] introduced the following modified signed likelihood ratio test statistic:

$$w_s^* = w_s + w_s^{-1} \log \varepsilon, \quad (9)$$

where

$$\varepsilon = |\hat{J}|^{1/2} |\tilde{U}'|^{-1} |\tilde{J}_{\psi\psi}|^{1/2} \frac{w_s}{[(\hat{\ell}' - \tilde{\ell}')^\top (\tilde{U}')^{-1}]_\kappa},$$

U being the score function and J being the matrix of second-order derivatives of the log-likelihood function. Notice that $-J$ is the observed information matrix. Here, $\tilde{U}'$ and $\tilde{\ell}'$ are, respectively, the score vector and the log-likelihood derivative with respect to $\hat{\theta}$, both evaluated at $\hat{\theta}$. Similarly, $\hat{\ell}'$ denotes the log-likelihood derivative with respect to $\hat{\theta}$ evaluated at $\hat{\theta}$. Additionally, $J_{\psi\psi}$ denotes the $s \times s$ matrix of second derivatives of the log-likelihood function with respect to ψ and $[(\hat{\ell}' - \tilde{\ell}')^\top (\tilde{U}')^{-1}]_\kappa$ is the element of vector $(\hat{\ell}' - \tilde{\ell}')^\top (\tilde{U}')^{-1}$ corresponding to the parameter κ , which in this case is scalar. It is noteworthy that the derivation of such quantities is oftentimes quite cumbersome. The proposed test statistic is standard normally distributed under the null hypothesis with error of order $n^{-3/2}$.

Skovgaard [32] obtained approximations to the sample space derivatives required to compute the test statistic given in (9) when the parameter of interest is scalar. His results were later generalized for vector-valued parameter of interest [33].

Several authors have recently obtained that adjustment for different classes of models; see, e.g. [16,26,30]. In particular, we note that the Skovgaard adjustment was derived by Ferrari and Pinheiro [17] for performing testing inferences in the beta regression model. In what follows, we shall derive a similar adjustment for likelihood ratio testing inference in the unit gamma regression model.

By approximating $\tilde{U}'$, $\tilde{\ell}'$ and $\hat{\ell}'$, Skovgaard [33] obtained the following adjusted likelihood ratio statistic:

$$w^* = w - 2 \log \xi,$$

where

$$\xi = \frac{\{|\tilde{I}||\hat{I}|\tilde{J}_{\psi\psi}|\}^{1/2}}{|\tilde{I}||\{\tilde{I}\tilde{\Upsilon}^{-1}\hat{I}^{-1}\tilde{\Upsilon}\}_{\psi\psi}|^{1/2}} \frac{\{\tilde{U}^\top \tilde{\Upsilon}^{-1} \hat{I}^{-1} \tilde{\Upsilon} \tilde{I}^{-1} \tilde{U}\}^{l/2}}{w^{l/2-1} \tilde{U}^\top \tilde{\Upsilon}^{-1} \tilde{q}}.$$

As noted earlier, when the relevant regularity conditions are satisfied, the negative Hessian, $-J_{\psi\psi}$, is the observed information matrix relative to ψ . Note that $\tilde{q}$ is a vector of dimension $l+s$ and $\tilde{\Upsilon}$ is a matrix of dimension $(l+s) \times (l+s)$. Under $\mathcal{H}_0$, w^* is asymptotically distributed as χ_l^2 . The quantities $\tilde{q}$ and $\tilde{\Upsilon}$ come from

$$q = \mathbb{E}[U(\theta_1)(\ell(\theta_1) - \ell(\theta))] \quad \text{and} \quad \Upsilon = \mathbb{E}[U(\theta_1)U^\top(\theta)]$$

by replacing θ_1 with $\hat{\theta}$ and θ with $\tilde{\theta}$ after the expected values are computed. Here, $\hat{\theta}$ and $\tilde{\theta}$ denote, respectively, the unrestricted and restricted maximum likelihood estimators of θ .

A test statistic which is asymptotically equivalent to w^* is

$$w^{**} = w \left(1 - \frac{1}{w} \log \xi \right)^2.$$

A clear advantage of w^{**} is that it is always nonnegative. Under $\mathcal{H}_0$, both test statistics are χ_l^2 distributed with high degree of accuracy. The adjusted statistics are invariant under reparameterizations of the form $(\kappa, \psi) \rightarrow (\kappa, \varphi(\kappa, \psi))$.

In what follows, we shall obtain closed form expressions for the quantities $\tilde{q}$ and $\tilde{\Upsilon}$ in the class of unit gamma regression models. After some algebra, we obtain the following

closed-form expressions for $\bar{q}$ and $\bar{\Upsilon}$:

$$\bar{q} = \begin{bmatrix} X^\top \hat{T}_1 \hat{\Phi}^{-1} \hat{\mathcal{M}}^{-1} \hat{D}(\mathcal{I} + \hat{D}) \{ \hat{V}^*(\hat{D} - \tilde{D}) + \hat{C}(\hat{\Phi} - \tilde{\Phi}) \}_\iota \\ H^\top \hat{T}_2 \{ (\hat{P} \hat{V}^* + \hat{C})(\hat{D} - \tilde{D}) + (\hat{P} \hat{C} + \hat{V}^\dagger)(\hat{\Phi} - \tilde{\Phi}) \}_\iota \end{bmatrix}$$

and

$$\bar{\Upsilon} = \begin{bmatrix} X^\top \hat{T}_1 \hat{\Phi}^{-1} \hat{\mathcal{M}}^{-1} \hat{D}(\mathcal{I} + \hat{D}) \hat{V}^* & X^\top \hat{T}_1 \hat{\Phi}^{-1} \hat{\mathcal{M}}^{-1} \hat{D}(\mathcal{I} + \hat{D}) \\ \times (\mathcal{I} + \tilde{D}) \tilde{D} \tilde{\mathcal{M}}^{-1} \tilde{\Phi}^{-1} \tilde{T}_1 X & \times \{ \hat{V}^* \tilde{P} + \hat{C} \} \tilde{T}_2 H \\ H^\top \hat{T}_2 \{ \hat{P} \hat{V}^* + \hat{C} \} (\mathcal{I} + \tilde{D}) \tilde{D} \tilde{\mathcal{M}}^{-1} \tilde{\Phi}^{-1} \tilde{T}_1 X & H^\top \hat{T}_2 \{ \hat{P} \hat{V}^* \tilde{P} + (\hat{P} + \tilde{P}) \hat{C} + \hat{V}^\dagger \} \tilde{T}_2 H \end{bmatrix},$$

where $V^* = \text{diag}(v_1^*, \dots, v_n^*)$, $V^\dagger = \text{diag}(v_1^\dagger, \dots, v_n^\dagger)$, $C = \text{diag}(c_1^*, \dots, c_n^*)$. Details on the derivation of such results can be found in the Appendix and also in Guedes [20].

We emphasize that the two adjusted likelihood ratio statistics can be easily computed using standard software and computing environments since such computation only entails basic matrix operations.

4. Numerical evidence

In what follows, we shall report the results of Monte Carlo simulations that were performed to evaluate the finite sample performances of the standard likelihood ratio test (w) and its two corrected counterparts (w^* and w^{**}) in unit gamma regression models. Parameter estimation was carried out by numerically maximizing the log-likelihood function using the BFGS quasi-Newton optimization algorithm with analytic first derivatives. The number of Monte Carlo replications is 10,000, i.e. all reported results are based on 10,000 samples. All simulations were performed using the Ox matrix programming language [12].

At the outset, we consider the fixed dispersion unit gamma regression model with mean submodel given by

$$\log \left(\frac{\mu_i}{1 - \mu_i} \right) = \beta_1 + \beta_2 x_{i2} + \beta_3 x_{i3} + \beta_4 x_{i4}.$$

The covariates values are obtained as random draws from the standard uniform distribution, i.e. from $\mathcal{U}(0, 1)$. We consider three different ranges of values for μ : (i) close to zero (μ_1), (ii) around 0.5 (μ_2) and (iii) close to one (μ_3). To that end, we use different parameter values in the data generating mechanism. There are two scenarios. In the first one, we test $\mathcal{H}_0 : \beta_4 = 0$. Hence, $l = 1$ (one restriction). The parameter values for the three ranges of μ are: (i) $\beta_1 = -1.5$, $\beta_2 = -1.5$, $\beta_3 = 1.2$ and $\beta_4 = 0$, (ii) $\beta_1 = -1.5$, $\beta_2 = 1.5$, $\beta_3 = 1.2$ and $\beta_4 = 0$, (iii) $\beta_1 = 1.5$, $\beta_2 = 1.5$, $\beta_3 = 1.2$ and $\beta_4 = 0$. In the second scenario, we test $\mathcal{H}_0 : \beta_3 = \beta_4 = 0$. Hence, $l = 2$ (two restrictions). The parameter values were specified as: (i) $\beta_1 = -1.5$, $\beta_2 = -1.5$ and $\beta_3 = \beta_4 = 0$, (ii) $\beta_1 = -1.5$, $\beta_2 = 1.5$ and $\beta_3 = \beta_4 = 0$, (iii) $\beta_1 = 1.5$, $\beta_2 = 1.5$ and $\beta_3 = \beta_4 = 0$. In both scenarios, the null hypotheses are tested against bilateral alternative hypotheses. We consider three values for the precision parameter, namely: $\phi = 5, 10, 30$. The sample sizes are $n = 20, 40, 60$. The tests significance levels are $\alpha = 10\%$, $\alpha = 5\%$ and $\alpha = 1\%$.

The tests null rejection rates are presented in Table 1 (entries are percentages). The reported results lead to interesting conclusions. First, the likelihood ratio test is considerably oversized (liberal) when the sample size is small. For instance, when $n = 20$, $\phi = 10$

Table 1. Null rejection rates (%), fixed dispersion I.

ϕ	l	n	μ_1			μ_2			μ_3		
			w	w^*	w^{**}	w	w^*	w^{**}	w	w^*	w^{**}
$\alpha = 10\%$											
30	1	20	14.9	10.0	10.3	14.7	10.0	10.4	14.7	10.0	10.4
		40	12.2	9.8	9.9	11.8	9.8	9.9	12.1	9.9	10.0
		60	11.3	9.9	9.9	11.3	9.8	9.9	11.1	9.8	9.9
	2	20	16.2	10.3	10.7	16.3	10.5	10.8	16.2	10.4	10.8
		40	13.1	10.0	10.2	13.0	10.2	10.3	12.9	10.2	10.3
		60	11.8	10.1	10.1	11.8	10.1	10.2	11.8	10.1	10.2
10	1	20	14.9	9.9	10.3	14.8	9.9	10.3	14.7	9.7	10.1
		40	12.0	9.9	9.9	11.7	9.6	9.7	11.9	9.7	11.9
		60	11.9	10.4	10.5	11.8	10.6	10.7	11.7	10.5	10.5
	2	20	15.6	9.7	10.1	15.8	10.0	10.3	15.5	9.9	10.1
		40	12.6	9.8	9.9	12.3	9.7	9.7	12.3	9.6	9.7
		60	11.8	10.1	10.1	11.5	10.0	10.0	11.4	10.1	10.1
5	1	20	15.0	10.4	10.7	15.0	10.4	10.8	14.7	10.3	10.6
		40	11.6	9.3	9.5	11.7	9.7	9.8	11.2	9.3	9.4
		60	11.3	9.8	10.0	11.4	10.0	10.1	11.2	9.8	9.9
	2	20	16.5	9.8	10.2	15.8	9.6	10.0	15.7	9.8	10.1
		40	12.7	10.0	10.0	12.6	10.0	10.0	12.3	9.9	10.0
		60	11.2	9.5	9.6	11.5	9.9	9.9	11.2	9.9	9.9
$\alpha = 5\%$											
30	1	20	8.4	5.2	5.4	8.6	5.2	5.4	8.2	5.1	5.3
		40	6.2	4.7	4.9	6.4	4.8	4.9	6.3	4.7	4.8
		60	6.0	5.2	5.2	6.2	5.1	5.2	5.8	5.1	5.2
	2	20	9.5	5.0	5.2	9.5	5.3	5.4	9.5	5.1	5.4
		40	6.7	5.1	5.2	6.7	4.7	4.8	6.7	4.8	4.9
		60	6.4	5.0	5.0	6.2	5.0	5.0	6.1	5.1	5.1
10	1	20	8.3	5.3	5.5	8.4	5.1	5.3	8.1	5.1	5.4
		40	6.4	4.8	4.9	6.0	4.6	4.7	6.1	4.6	4.7
		60	6.3	5.3	5.4	6.0	5.1	5.2	6.2	5.3	5.3
	2	20	9.2	5.2	5.4	9.1	5.1	5.3	8.9	5.2	5.3
		40	6.6	4.8	4.8	6.5	4.9	4.9	6.4	4.7	4.8
		60	6.4	5.0	5.1	6.2	5.0	5.0	6.3	5.2	5.2
5	1	20	8.9	5.3	5.6	8.5	5.1	5.4	8.5	5.1	5.3
		40	6.3	5.0	5.1	6.2	4.8	4.9	6.1	4.9	4.9
		60	5.6	4.8	4.9	5.9	5.0	5.2	5.6	4.8	4.8
	2	20	9.4	4.6	4.7	9.0	4.8	5.1	8.8	4.8	4.9
		40	6.9	4.8	4.8	6.6	4.9	4.9	6.6	4.8	4.8
		60	6.0	4.7	4.8	5.9	4.7	4.7	5.9	4.8	4.8
$\alpha = 1\%$											
30	1	20	2.4	1.0	1.1	2.4	1.0	1.0	2.3	0.9	1.0
		40	1.3	0.9	0.9	1.4	1.0	1.0	1.4	0.9	1.0
		60	1.4	1.0	1.0	1.4	1.0	1.0	1.3	1.0	1.1
	2	20	2.5	1.0	1.1	2.6	0.9	1.0	2.5	0.9	0.9
		40	1.6	0.9	1.0	1.5	1.0	1.0	1.5	1.0	1.0
		60	1.3	1.0	1.0	1.4	1.0	1.0	1.4	1.0	1.0
10	1	20	2.4	1.0	1.1	2.4	1.0	1.2	2.3	1.0	1.2
		40	1.4	0.9	1.0	1.3	0.9	0.9	1.4	0.9	0.9
		60	1.4	1.0	1.1	1.2	0.9	1.0	1.3	0.9	1.0
	2	20	2.8	1.0	1.1	2.7	1.0	1.1	2.7	1.0	1.1
		40	1.5	0.9	0.9	1.5	1.0	1.0	1.4	0.9	0.9
		60	1.4	1.0	1.0	1.5	1.0	1.0	1.4	1.1	1.1
5	1	20	2.4	1.0	1.1	2.4	1.0	1.1	2.2	1.0	1.1
		40	1.4	0.9	1.0	1.4	0.9	1.0	1.4	0.9	1.0
		60	1.3	1.0	1.1	1.3	1.0	1.1	1.4	1.0	1.0
	2	20	2.6	0.9	1.0	2.5	1.0	1.0	2.4	0.9	1.0
		40	1.4	0.8	0.8	1.5	0.8	0.9	1.3	0.8	0.9
		60	1.4	0.9	0.9	1.2	0.9	0.9	1.3	0.9	0.9

and $l = 2$, the test null rejection rate exceeds 15% at the 10% significance level for all three ranges of mean values. Second, the two corrected tests (w^* and w^{**}) perform considerably better than the standard likelihood ratio test. Under the same conditions ($n = 20$, $\phi = 10$, $l = 2$, 10% significance level), their null rejection rates range from 9.7% to 10% (w^*) and from 10.1% to 10.3% (w^{**}) across the different ranges of mean values. Third, the two corrected tests behave similarly, w^* being slightly more conservative than w^{**} . Overall, they perform very well, i.e. they are quite effective when it comes to controlling the frequency of type I error. Fourth, the range of values for the mean responses do not noticeably affect the tests performances. Fifth, the impact of the precision parameter value on the tests finite sample performances is small.

The second set of simulations is based on the following fixed dispersion unit gamma regression model:

$$\log\left(\frac{\mu_i}{1 - \mu_i}\right) = \beta_1 + \sum_{j=2}^p \beta_j x_{ij}.$$

The interest lies in testing $\mathcal{H}_0 : \beta_2 = 0$ against $\mathcal{H}_1 : \beta_2 \neq 0$. We shall report numerical results for $p = 3, 4, 5$. Notice that the number of nuisance parameters increases with p . The sample sizes are $n = 20, 40, 60$, $\phi = 30$ and, as before, we consider three ranges of values for μ . The tests null rejection rates can be found in Table 2 (entries are percentages). The figures in this table show that the likelihood ratio test tends to become more liberal as the number of nuisance parameters increases. Consider, e.g. $n = 20$ and μ_2 (i.e. response means in the middle of the standard unit interval). At the 5% significance level, the test null rejection rates corresponding to $p = 3, 4, 5$ are, respectively, 7.0%, 8.4% and 9.4%. In contrast, the finite sample behaviors of the two corrected tests are much less affected by the increase in the number of nuisance parameters. Under the same conditions, for instance, the null rejection rates of w^* (w^{**}) when $p = 3, 4, 5$ are 4.7%, 4.9% and 5.1% (4.9%, 5.1% and 5.5%), respectively. It is also noteworthy that once again the two corrected tests behave similarly, w^* being slightly more conservative than w^{**} . As in the previous set of simulations, the range of values for the mean responses has no noticeable impact on the tests performances.

Figure 1 contains quantile-quantile (QQ) plots for the three test statistics. The sample size is $n = 20$, $\phi = 5, 10$ and the response means assume values in the middle of the standard unit interval. In each panel we plot the test statistics empirical quantiles against the corresponding asymptotic quantiles, i.e. against χ_1^2 quantiles. There is good agreement between the exact and asymptotic null distributions when the curve is close to the 45 degree line (the thin solid, diagonal line). Visual inspection of the QQ plots shows that the null distributions of the two corrected test statistics are much better approximated by the reference χ^2 distribution than that of the likelihood ratio test statistic. It is also clear that the corrected test statistics are less affected by the increase in the number of nuisance parameters.

We now consider the varying precision unit gamma regression model given by

$$\begin{aligned} \log\left(\frac{\mu_i}{1 - \mu_i}\right) &= \beta_1 + \beta_2 x_{i2} + \beta_3 x_{i3} + \beta_4 x_{i4}, \\ \log(\phi_i) &= \delta_1 + \delta_2 h_{i2} + \delta_3 h_{i3} + \delta_4 h_{i4}. \end{aligned}$$

Table 2. Null rejection rates (%), fixed dispersion II.

$\phi = 30$	w									w^*									w^{**}								
	μ_1			μ_2			μ_3			μ_1			μ_2			μ_3			μ_1			μ_2			μ_3		
	3	4	5	3	4	5	3	4	5	3	4	5	3	4	5	3	4	5	3	4	5	3	4	5	3	4	5
	$\alpha = 10\%$																										
$n = 20$	13.0	15.0	15.7	12.9	14.9	15.8	13.1	14.8	15.7	9.5	10.0	10.1	9.5	9.8	9.9	9.4	10.0	9.9	9.6	10.4	10.6	9.7	10.1	10.3	9.6	10.3	10.4
$n = 40$	11.3	12.8	13.0	11.1	12.6	12.8	11.1	12.6	13.1	9.5	10.6	10.2	9.7	10.4	10.3	9.5	10.3	10.3	9.6	10.7	10.4	9.8	10.5	10.4	9.7	10.3	10.5
$n = 60$	11.1	11.3	12.0	10.6	11.3	12.1	10.8	11.3	12.0	9.9	9.8	10.2	9.7	10.0	10.5	9.7	10.0	10.2	9.9	9.9	10.4	9.8	10.1	10.5	9.8	10.0	10.3
	$\alpha = 5\%$																										
$n = 20$	7.2	8.7	9.7	7.0	8.4	9.4	7.0	8.4	9.3	4.8	5.0	5.1	4.7	4.9	5.1	4.7	4.9	4.9	4.9	5.2	5.4	4.9	5.1	5.5	4.8	5.1	5.3
$n = 40$	6.0	6.9	7.2	5.9	6.7	7.1	6.0	6.8	7.3	4.9	5.4	5.3	4.8	5.4	5.4	4.8	5.2	5.4	5.0	5.4	5.4	4.8	5.4	5.4	4.9	5.3	5.6
$n = 60$	5.4	5.8	6.4	5.6	5.8	6.4	5.5	5.7	6.3	4.8	4.8	5.1	4.9	4.9	5.0	4.7	4.9	5.0	4.9	4.9	5.2	4.9	4.9	5.1	4.8	5.0	5.0
	$\alpha = 1\%$																										
$n = 20$	1.8	2.2	2.9	1.8	2.1	2.7	1.8	2.2	2.7	0.9	1.0	0.9	0.8	0.9	0.9	0.8	0.8	1.0	1.0	1.1	1.0	0.9	1.0	1.1	0.9	0.9	1.1
$n = 40$	1.4	1.7	1.9	1.4	1.7	1.8	1.4	1.6	1.8	1.0	1.1	1.1	1.0	1.1	1.1	1.0	1.0	1.1	1.1	1.1	1.1	1.0	1.1	1.1	1.1	1.1	1.2
$n = 60$	1.2	1.3	1.5	1.2	1.3	1.4	1.2	1.3	1.3	0.9	1.0	1.0	1.0	1.0	0.9	0.9	0.9	1.0	1.0	1.1	1.1	1.0	1.0	1.0	1.0	0.9	1.0

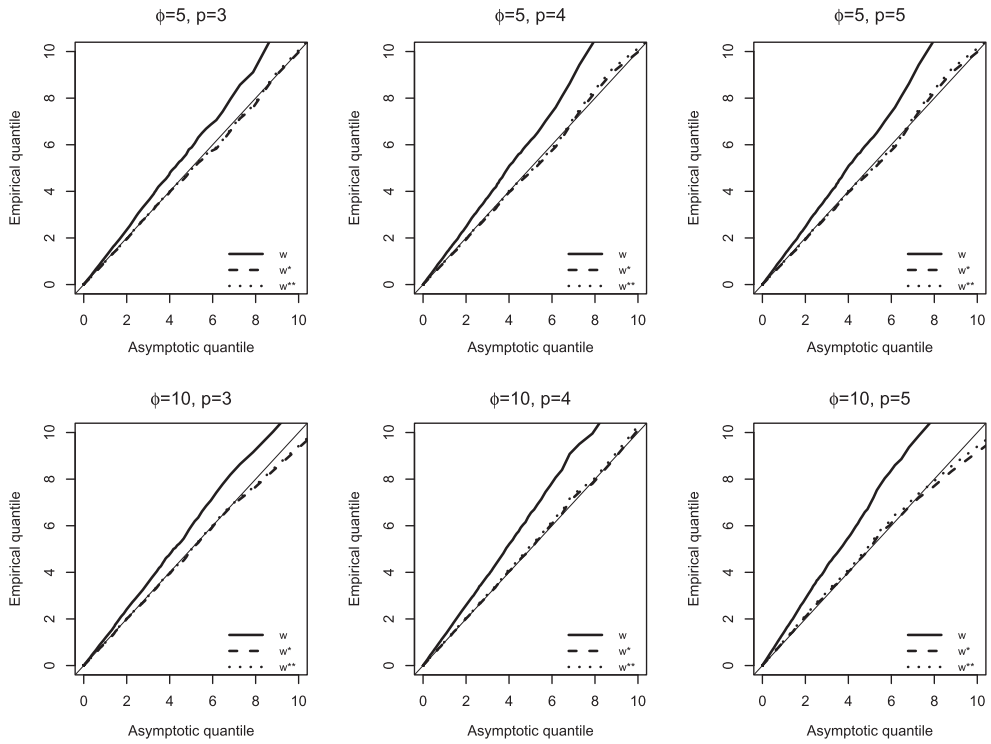


Figure 1. Quantile-quantile (QQ) plots, fixed dispersion, $l = 1, n = 20$.

Again, the values of all covariates are obtained as random standard uniform draws. We shall consider two separate scenarios. In the first scenario, we test $\mathcal{H}_0 : \delta_2 = \delta_3 = \delta_4 = 0$ ($l = 3$) against a two-sided alternative hypothesis. Notice that we test the null hypothesis of constant dispersion against the alternative hypothesis of varying dispersion. The parameter values used for data generation are $\beta_1 = 1.5, \beta_2 = 1.5, \beta_3 = 4.5, \beta_4 = -3.5, \delta_2 = \delta_3 = \delta_4 = 0, \delta_1 = \log(30), \log(10), \log(5)$. In the second scenario, we test $\mathcal{H}_0 : \beta_3 = \beta_4 = 0, \delta_2 = \delta_3 = \delta_4 = 0$ ($l = 5$). Data generation was carried out using $\beta_1 = 1.5, \beta_2 = 1.5, \beta_3 = 0, \beta_4 = 0, \delta_1 = (\log(30), \log(10), \log(5)), \delta_2 = \delta_3 = \delta_4 = 0$. The sample sizes are $n = 20, 30, 40, 50, 60$ in the first scenario and $n = 20, 40, 60$ in the second scenario. The results (tests null rejection rates) are presented in Table 3 (entries are percentages).

The figures in Table 3 show that the likelihood ratio test tends to be considerably liberal in small samples, i.e. it tends to overreject the null hypothesis when it is true. For instance, when $n = 20, \delta_1 = \log(10)$ and $\alpha = 10\%$, the test null rejection rate equals 33.2%, i.e. it is over three times larger than the test significance level. When $n = 50$, it equals 15.1%; it still is over 50% larger than the nominal level. The corrected tests are less size distorted. Under the same conditions, e.g. the null rejection rates of w^* (w^{**}) are 12.9% and 15.7% for $n = 20$. It is noteworthy that w^* is less oversized than w^{**} , especially when n is small.

The tests null rejection rates for the second scenario (joint test on the parameters of both submodels) are presented in Table 4 (entries are percentages). Once again, the likelihood ratio test is substantially oversized in small samples. For instance, when $n = 20, \delta_1 = \log(30)$ and $\alpha = 10\%$, its null rejection rate equals 34.9%. The two corrected tests

Table 3. Null rejection rates (%), varying dispersion I.

δ_1	n	$\alpha = 10\%$			$\alpha = 5\%$			$\alpha = 1\%$		
		w	w^*	w^{**}	w	w^*	w^{**}	w	w^*	w^{**}
log(30)	20	34.8	12.5	15.6	24.5	7.0	9.0	10.8	1.6	2.3
	30	22.5	10.6	11.8	14.2	5.8	6.4	4.8	1.1	1.2
	40	17.6	9.7	10.2	10.4	5.0	5.3	2.9	1.0	1.0
	50	15.7	9.9	10.2	8.6	4.7	4.9	2.3	0.9	1.0
	60	14.6	10.2	10.3	8.0	5.1	5.2	2.1	1.0	1.1
log(10)	20	33.2	12.9	15.7	23.5	7.2	9.1	10.2	1.7	2.4
	30	22.4	11.4	12.3	14.5	5.8	6.5	4.5	1.2	1.4
	40	17.6	10.7	11.1	10.8	5.2	5.4	3.2	1.2	1.2
	50	15.1	10.1	10.2	8.6	5.1	5.2	2.3	1.0	1.1
	60	13.9	9.6	9.7	7.7	4.8	4.8	1.8	0.8	0.9
log(5)	20	32.5	12.5	15.6	22.7	7.1	9.2	10.0	2.1	2.7
	30	22.2	11.6	12.6	14.0	6.1	6.6	4.7	1.4	1.6
	40	17.7	10.8	11.2	10.4	5.5	5.7	2.8	1.0	1.1
	50	14.7	9.7	9.9	8.4	4.9	5.0	2.0	1.0	1.1
	60	14.4	10.5	10.6	8.2	5.5	5.6	2.2	1.1	1.2

Table 4. Null rejection rates (%), varying dispersion II.

δ_1	n	$\alpha = 10\%$			$\alpha = 5\%$			$\alpha = 1\%$		
		w	w^*	w^{**}	w	w^*	w^{**}	w	w^*	w^{**}
log(30)	20	34.9	8.8	10.5	24.6	5.5	6.5	10.3	2.1	2.6
	40	17.7	8.9	9.4	10.3	4.1	4.4	3.0	0.8	0.9
	60	14.7	9.4	9.7	7.9	4.4	4.6	1.8	0.9	0.9
log(10)	20	34.1	8.2	9.9	23.8	4.8	5.8	9.8	1.8	2.3
	40	17.5	8.7	9.3	10.3	4.4	4.7	2.8	0.9	1.0
	60	13.9	9.1	9.2	7.8	4.6	4.7	1.9	0.9	0.9
log(5)	20	34.1	8.2	9.9	23.8	4.8	5.8	9.8	1.8	2.3
	40	17.5	8.7	9.3	10.3	4.4	4.7	2.8	0.9	1.0
	60	13.9	9.1	9.2	7.8	4.6	4.7	1.9	0.9	0.9

perform well. Under the same circumstance, their null rejection rates are 8.8% (w^*) and 10.5% (w^{**}). As expected, the tests performances improve as the sample size increases.

Figure 2 contains QQ plots for the second simulation design. It is clear that the null distribution of w is poorly approximated by the limiting χ^2 distribution. It is also clear that such an approximation is much more precise when used with the two corrected test statistics derived in this paper, especially when $n \geq 40$. Notice that the case in which there are only 20 observations in the sample is quite challenging since there are five restrictions under test. Even with $n = 20$, however, the χ^2 approximation is somewhat precise when used with w^* and w^{**} , except in the distribution upper tail.

The final set of simulations concerns a nonnested hypothesis test, i.e. the use of the likelihood ratio test (w) and its improved versions to distinguish between unit gamma regression models that are nonnested. We shall consider the J test outlined for beta regressions by Cribari-Neto and Lucena [9] and more generally for GAMLSS (generalized additive models for location, scale and shape) by Cribari-Neto and Lucena [10]. We refer readers to such articles for details. When there are two competing models, say Model $\mathcal{M}_1$ and Model $\mathcal{M}_2$, they are sequentially tested against the other, i.e. one tests Model $\mathcal{M}_1$ against Model $\mathcal{M}_2$ and then tests Model $\mathcal{M}_2$ against Model $\mathcal{M}_1$. It is thus possible to accept one model as the true model and reject the alternative model, accept both models (i.e. conclude that

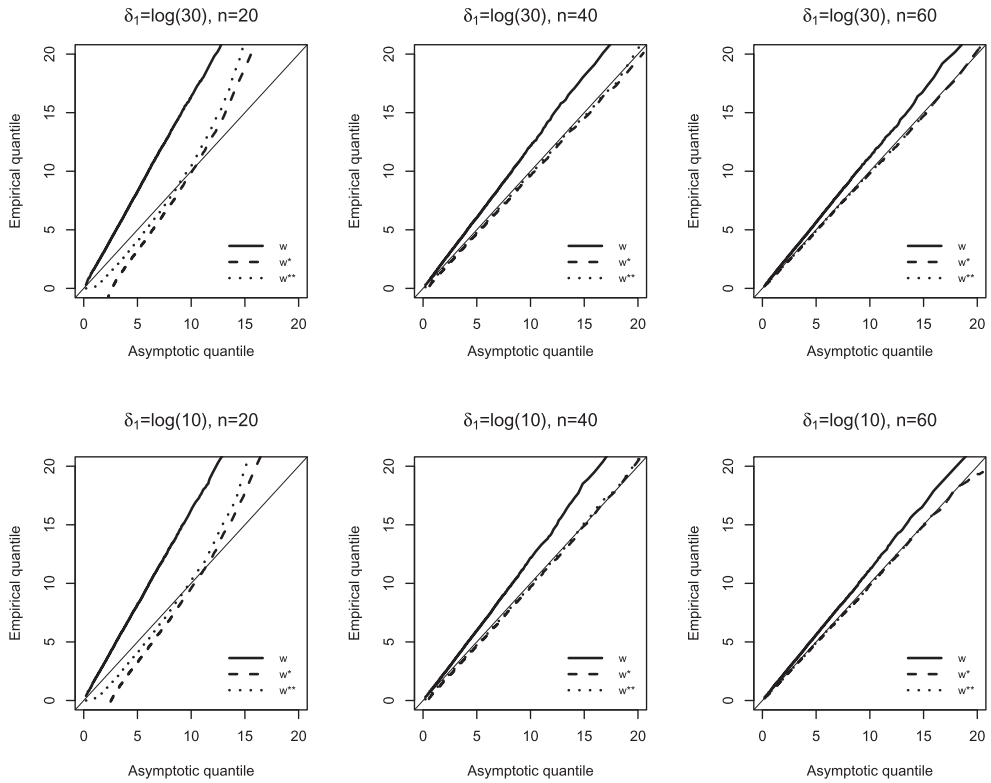


Figure 2. Quantile-quantile (QQ) plots, varying dispersion, $l = 5$.

the two models are empirically indistinguishable) or reject both models. In the simulations, we shall consider two alternative fixed dispersion unit gamma regression models which are evaluated under two different scenarios. In the first scenario, they differ in the link functions used (logit and log-log) whereas in the second scenario they differ in the link functions used (logit and log-log) and also in one of the covariates. The sample size is $n = 40$, the values of the precision parameter are $\phi = 5, 10, 15, 20, 30$ and the values of all covariates were selected as random $\mathcal{U}(0, 1)$ draws. The true data generating mechanism is Model $\mathcal{M}_1$ so that the rejection rates of the test of that model against Model $\mathcal{M}_2$ are null rejection rates (i.e. size) whereas those of the test of Model $\mathcal{M}_2$ versus the true model are nonnull rejection rates (power).

In the first scenario the two models only differ in the link functions used: logit ($\mathcal{M}_1$) and log-log ($\mathcal{M}_1$). The competing models are

$$\mathcal{M}_1 : \log\left(\frac{\mu_i}{1 - \mu_i}\right) = \beta_1 + \beta_2 x_{i2} + \beta_3 x_{i3},$$

$$\mathcal{M}_2 : -\log(-\log(\mu_i)) = \beta_1 + \beta_2 x_{i2} + \beta_3 x_{i3}.$$

The parameter values used for data generation are $\beta_1 = 1.1$, $\beta_2 = 3.8$ and $\beta_3 = -4.2$ which result in mean response values in $(0.12, 0.98)$.

Table 5. Null and nonnull rejection rates (%), J test, first scenario (models use different link functions).

true model: $\log\left(\frac{\mu_i}{1-\mu_i}\right) = \beta_1 + \beta_2 x_{i2} + \beta_3 x_{i3}$

ϕ	$\alpha = 10\%$			$\alpha = 5\%$			$\alpha = 1\%$		
	w	w^*	w^{**}	w	w^*	w^{**}	w	w^*	w^{**}
null rejection rates (size)									
30	11.5	9.4	9.4	6.1	4.6	4.7	1.4	1.0	1.0
20	12.3	9.9	10.0	6.4	5.1	5.1	1.4	0.9	0.9
15	12.2	9.9	10.0	6.5	4.8	4.9	1.5	1.0	1.0
10	12.2	10.1	10.1	6.4	4.8	4.9	1.3	0.8	0.9
5	11.7	9.4	9.5	6.0	4.6	4.7	1.4	1.0	1.1
nonnull rejection rates (power)									
30	97.8	97.1	97.1	95.1	93.4	93.5	83.1	77.6	77.8
20	91.8	90.0	90.0	85.2	81.8	82.0	64.7	57.7	58.0
15	84.5	81.4	81.6	74.7	70.5	70.7	50.7	43.9	44.1
10	71.8	67.9	68.0	59.3	54.1	54.2	34.2	27.9	28.0
5	49.9	45.6	45.8	37.5	32.4	32.5	17.1	12.9	13.1

The tests null and nonnull rejection rates (i.e. estimated sizes and powers) are presented in Table 5 (entries are percentages). At the outset, notice that the two improved tests (w^* and w^{**}) are nearly free of size distortions, even when the precision is small. For instance, when $\phi = 5$ (the smallest precision) and at the 10% nominal level the null rejection rates of w^* and w^{**} are, respectively, 9.4% and 9.5%. The estimated size of the standard likelihood ratio test (w) under the same conditions is 11.7%. It is also important to examine the tests powers, i.e. the tests ability to detect that the null hypothesis is false. It is noteworthy that even though the value of the precision parameter does not noticeably impact the tests sizes, it has a sizable impact on the tests powers; the smaller ϕ , the less powerful the tests. We also note that the powers of w are slightly higher than those of w^* and w^{**} which follows from the standard likelihood ratio test being slightly liberal, i.e. oversized. Additionally, when $\phi \geq 15$, all powers are above 80%.

In the second scenario, in addition to the difference in the link functions, each model includes a covariate that is not present in the other model. The competing models are

$$\mathcal{M}_1 : \log\left(\frac{\mu_i}{1-\mu_i}\right) = \beta_1 + \beta_2 x_{i2} + \beta_3 x_{i3},$$

$$\mathcal{M}_2 : -\log(-\log(\mu_i)) = \beta_1 + \beta_2 x_{i2} + \beta_4 x_{i4}.$$

The data were generated using $\beta_1 = 1.1$, $\beta_2 = 2.0$ and $\beta_3 = -1.1$, and the mean response assumes values in (0.60, 0.95).

The results (null and nonnull rejection rates expressed as percentages) for the second scenario are presented in Table 6. Again, w is slightly liberal and w^* and w^{**} are nearly free of size distortions, even when precision is small. For instance, when $\phi = 5$ and at the 5% significance level, the null rejection rates of these tests are, respectively, 7.0%, 5.7% and 5.7%. It is noteworthy that the three tests are now much more powerful than under the previous setup. For example, at the 10% significance level all nonnull rejection rates are in excess of 93%.

Table 6. Null and nonnull rejection rates (%), J test, second scenario (models use different link functions and also differ by a covariate).

true model: $\log\left(\frac{\mu_i}{1-\mu_i}\right) = \beta_1 + \beta_2 x_{i2} + \beta_3 x_{i3}$									
ϕ	$\alpha = 10\%$			$\alpha = 5\%$			$\alpha = 1\%$		
	w	w^*	w^{**}	w	w^*	w^{**}	w	w^*	w^{**}
null rejection rates (size)									
30	11.9	10.0	10.2	6.1	4.9	5.0	1.5	1.0	1.1
20	11.9	9.8	9.9	6.2	4.9	5.0	1.4	0.8	0.8
15	11.3	9.5	9.6	6.0	4.6	4.7	1.4	0.9	0.9
10	11.2	9.4	9.5	5.8	4.6	4.6	1.3	0.8	0.8
5	12.4	10.9	10.9	7.0	5.7	5.7	1.5	1.1	1.1
nonnull rejection rates (power)									
30	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0
20	100.0	100.0	100.0	99.9	99.9	99.9	99.9	99.9	99.9
15	100.0	100.0	100.0	99.9	99.9	99.9	99.5	99.3	99.3
10	99.6	99.5	99.5	99.3	99.0	99.0	96.9	95.7	95.7
5	94.4	93.5	93.6	90.6	89.0	89.0	77.4	73.5	73.7

Overall, the numerical evidence in this section shows that likelihood ratio inferences in unit gamma regressions can be quite unreliable in small samples. Much more precise inferences can be achieved by using the two corrected test statistics we derived. We also note that testing inferences based on w^* tend to be slightly more accurate than those based on w^{**} .

5. Empirical applications

In what follows we shall present two empirical applications that use both the unit gamma and the beta regression model.

5.1. First application

The data were analyzed by Smithson and Verkuilen [34] and contain 44 observations on reading accuracy of dyslexic and nondyslexic Australian children. The variable of interest (y) are reading accuracy indices of such children. The independent variables are: dyslexia versus nondyslexia status (x_2) and nonverbal IQ converted to z -scores (x_3). We also consider two additional covariates, namely: an interaction variable ($x_2 \times x_3$) and z -scores squared (x_3^2). The participants (19 dyslexics and 25 controls) were students from primary schools in the Australian Capital Territory. The ages of the 44 children range from eight years five months to twelve years three months. The covariate x_2 is a dummy variable, which equals 1 if the child is dyslexic and -1 otherwise. The observed scores were linearly transformed from their original scale to the open unit interval $(0, 1)$; see Smithson and Verkuilen [34]. These data were also analyzed by Espinheira *et al.* [14], Grün *et al.* [19], Cribari-Neto and Queiroz [11] and Bayer and Cribari-Neto [5] using beta regression models. It is noteworthy that different regression models were used in the literature. One of such models uses the interaction variable ($x_2 \times x_3$) and the squared values of the z -scores in the precision submodel. According to Bayer and Cribari-Neto [5], this is the model selected by most standard beta regression model selection approaches. We shall denote such a model

by BetaModel–I. It is given by

$$\log\left(\frac{\mu_i}{1 - \mu_i}\right) = \beta_1 + \beta_2 x_{i2} + \beta_3 x_{i3}^2 + \beta_4 (x_{i2} \times x_{i3}^2), \quad (10)$$

$$\log(\phi_i) = \delta_1 + \delta_2 x_{i2} + \delta_3 x_{i3} + \delta_4 x_{i3}^2 + \delta_5 (x_{i2} \times x_{i3}). \quad (11)$$

At the outset, we assume that the response (y) is beta distributed. As explained above, the interest lies in determining whether x_3^2 and $x_2 \times x_3$ should be used as precision covariates in BetaModel–I. To that end, we test the null hypothesis $\mathcal{H}_0 : \delta_4 = \delta_5 = 0$ against a two-sided alternative hypothesis. The likelihood ratio test statistic (w) equals 16.410 (p -value < 0.001). We also computed the adjusted test statistics w^* and w^{**} using the results in Ferrari and Pinheiro [17]; their values are, respectively, 5.544 (p -value = 0.063) and 7.342 (p -value = 0.025). All three tests reject the null hypothesis at the 10% significance level, our significance level of reference. We note, however, that only the standard likelihood ratio test rejects the null hypothesis at the 5% significance level; indeed, it does so even at the 1% significance level.

Next, we test $\mathcal{H}_0 : \delta_5 = 0$ against $\mathcal{H}_1 : \delta_5 \neq 0$. We obtain $w = 7.248$ (p -value = 0.007), $w^* = 1.067$ (p -value = 0.302) and $w^{**} = 2.385$ (p -value = 0.123). It is noteworthy that the conclusion yielded by the standard likelihood ratio test ($\mathcal{H}_0$ is rejected, even at the 5% significance level) is different from that reached using the corrected tests ($\mathcal{H}_0$ is not rejected).

Based on the results of two corrected tests, we removed the interaction variable ($x_2 \times x_3$) from the precision submodel, thus arriving at the following reduced model:

$$\log\left(\frac{\mu_i}{1 - \mu_i}\right) = \beta_1 + \beta_2 x_{i2} + \beta_3 x_{i3}^2 + \beta_4 (x_{i2} \times x_{i3}^2),$$

$$\log(\phi_i) = \delta_1 + \delta_2 x_{i2} + \delta_3 x_{i3} + \delta_4 x_{i3}^2.$$

We shall denote such a model by BetaModel–II. Its parameter estimates and standard errors are presented in Table 7.

We now test $\mathcal{H}_0 : \delta_4 = 0$ against a two-sided alternative hypothesis. The test statistics values are $w = 9.162$ (p -value = 0.002), $w^* = 5.159$ (p -value = 0.023) and $w^{**} = 5.596$ (p -value = 0.018). The three tests reject the null hypothesis, thus suggesting that x_3^2 should be included as a precision covariate.

Finally, we test $\mathcal{H}_0 : \delta_2 = \delta_3 = \delta_4 = 0$. We obtain $w = 33.689$ (p -value < 0.001), $w^* = 20.549$ (p -value < 0.001) and $w^{**} = 21.830$ (p -value < 0.001). All three tests reject the null hypothesis, even at the 1% significance level. Hence, we consider BetaModel–II to be the correct model based on which all further inferences should be made. Its pseudo- R_{FC}^2 (which is the square of the correlation coefficient between $g_1(y_i)$ and $\hat{\eta}_i$) equals 0.624. The alternative pseudo- R^2 measure proposed by Nagelkerke [28], which we denote by pseudo- R_N^2 , is considerably higher: 0.854.

Table 7. Parameter estimates and standard errors: BetaModel–II, dyslexia data.

Parameter	β_1	β_2	β_3	β_4	δ_1	δ_2	δ_3	δ_4
Estimate	1.116	−0.791	0.443	−0.411	2.593	1.249	1.050	0.897
S.E.	0.151	0.151	0.070	0.070	0.307	0.262	0.251	0.209

We shall now turn to unit gamma regression modeling, i.e. we shall assume that the response (y) follows the unit gamma law. At the outset, we consider the model defined by (10) and (11), which we denote by GammaModel-I. We test the null hypothesis $\mathcal{H}_0 : \delta_4 = \delta_5 = 0$ against a two-sided alternative hypothesis. The values of the likelihood ratio test statistic (w) and its two corrected variants (w^* and w^{**}) are, respectively, 6.876 (p -value = 0.032), 0.380 (p -value = 0.827) and 1.914 (p -value = 0.384). It is noteworthy that the testing inferences are quite different when based on the standard likelihood ratio test ($\mathcal{H}_0$ is rejected) and on the corrected tests ($\mathcal{H}_0$ is not rejected). Indeed, the p -value of w^* , the best performing test in our Monte Carlo simulations, is quite large (in excess of 0.8), which indicates that there is very little evidence against the null hypothesis.

Since the difference between the value of w and the values of w^* and w^{**} is relatively large, we performed a Monte Carlo simulation using the restricted model as the true data generating process to assess the performances of the corrected tests. The sample size is $n = 44$, the observed values of the regressors were used and the parameter values are taken to be the restricted parameter estimates, i.e. the data are generated under the null hypothesis. The number of replications is 10,000. The null rejection rates of w (w^*) [w^{**}] at the 10% and 5% significance levels are 24.6% and 16.4% (8.0% and 4.1%) [9.9% and 5.2%]. The likelihood ratio test is considerably liberal and the two corrected tests display much better control of the type I error probability, especially that based on w^{**} .

Based on the testing inference reached using the two corrected tests, we removed the covariates x_3^2 and $x_2 \times x_3$ from the precision submodel, thus arriving at the model we denote by GammaModel-II:

$$\log\left(\frac{\mu_i}{1 - \mu_i}\right) = \beta_1 + \beta_2 x_{i2} + \beta_3 x_{i3}^2 + \beta_4 (x_{i2} \times x_{i3}^2), \quad (12)$$

$$\log(\phi_i) = \delta_1 + \delta_2 x_{i2} + \delta_3 x_{i3}. \quad (13)$$

The parameter estimates and standard errors obtained for such a model are displayed in Table 8.

We now test the null hypothesis of fixed precision, i.e. we test $\mathcal{H}_0 : \delta_2 = \delta_3 = 0$. The test statistics values are $w = 47.920$, $w^* = 41.923$ and $w^{**} = 42.111$; all three p -values are quite small and, as a consequence, the null hypothesis is rejected by all three tests. There is, thus, evidence, of varying dispersion. We then take GammaModel-II to be the correct model. For that model, $\text{pseudo-}R_{FC}^2 = 0.607$ and $\text{pseudo-}R_N^2 = 0.856$.

The important question is: How do the selected beta and unit gamma regression model compare for the application at hand? At the outset, we note that the $\text{pseudo-}R^2$ of the beta regression model is slightly higher than that of the unit gamma model. This criterion thus favors the beta model. The difference in $\text{pseudo-}R^2$ is, however, quite minor (2.7% when $\text{pseudo-}R_{FC}^2$ is used and only 0.2% when $\text{pseudo-}R_N^2$ is used). Next, we compute, for both models, the values of three model selection criteria: AIC, AICc (which incorporates a

Table 8. Parameter estimates and standard errors: GammaModel-II, dyslexia data.

Parameter	β_1	β_2	β_3	β_4	δ_1	δ_2	δ_3
Estimate	1.062	-0.663	0.474	-0.476	1.603	2.030	0.625
S.E.	0.150	0.150	0.086	0.086	0.204	0.241	0.241

Table 9. Model selection criteria, dyslexia data.

Model	AIC	AICc	BIC
BetaModel-II	-121.51	-117.40	-107.24
GammaModel-II	-124.01	-120.90	-111.52

Table 10. J tests p -values, dyslexia data.

J test	Statistic		
	w	w^*	w^{**}
GammaModel-II versus BetaModel-II	0.0000	0.0866	0.0003
BetaModel-II versus GammaModel-II	0.0000	0.0004	0.0004

small sample correction) and BIC. These criteria are routinely used for comparing different regression and time series models. We do not intend to use them for comparing a large number of models. Instead, we shall use them to distinguish between the two regression models we arrived at using the modified (i.e. small sample adjusted) likelihood ratio tests. The values of the three criteria for the two models are presented in Table 9. All three criteria favor the unit gamma regression model. The difference in AIC (AICc) [BIC] between the two models, in favor of the unit gamma model, equals 2.50 (3.50) [4.28]. We note that according to Burnham and Anderson [7], the larger the difference between the models AIC the stronger the evidence in favor of that with the smallest AIC, and when such a difference exceeds 2 it is possible to conclude that there is less support for the model with larger AIC. The other two criteria (AICc and BIC) favor the unit gamma regression model as well.

We shall now compare the two regression models (GammaModel-II and BetaModel-II) by means of a hypothesis test. Since such models are nonnested, we shall use the J test; for details, see Cribari-Neto and Lucena [9,10]. Here, the J tests are performed using the standard likelihood ratio test statistic (w) and also the two corrected test statistics (w^* and w^{**}). The tests p -values are presented in Table 10. Based on the w^* test, we do not reject GammaModel-II and reject BetaModel-II at the 5% significance level. There is thus some evidence that the unit gamma regression model yields a better fit than the beta regression model for these data.

It is important to assess whether the inferences reached using J test are trustworthy. We reported simulation evidence on the J test in Section 4. Those simulations were performed using two unit gamma regression models and fixed dispersion, but now the two competing models have varying dispersion and belong to different classes of models (beta and unit gamma). We carried out a new set of simulations using $n = 44$, the observed covariate values (i.e. the covariate values from this application) and models GammaModel-II and BetaModel-II. Data generation was performed using the parameter estimates as the true parameter values. Again, the number of Monte Carlo replications is 10,000. For brevity, we only consider the best performing test (w^*) and one nominal level (5%). We take GammaModel-II as the true model and test it against BetaModel-II. The test's null rejection rate (size) is exactly 5.0% and the nonnull rejection rate (power) is 71.6%. Such numerical (Monte Carlo) evidence seems to suggest that the J test outcome reported above can be trusted.

Although slightly mixed, the overall evidence seems to favor the unit gamma regression model for the application at hand. Even though its pseudo- R^2 is slightly smaller than that

of the beta regression model, three model selection criteria and a nonnested hypothesis test single out the unit gamma model as the best fitting model.

The above empirical application shows the importance of performing accurate testing inferences when modeling data that assume values in the standard unit interval. When the sample size is small we recommend that such testing inferences be based on the two corrected tests introduced in this paper, especially on w^* .

We make available, as supplementary material, computer code that can be used to perform testing inferences, via w^* and w^{**} , on the parameters that index GammaModel-II.

5.2. Second application

The previous application showed the usefulness of the modified likelihood ratio tests considered in this paper. The beta and unit gamma models yielded similar data fits, the latter being marginally better at best. In the present application we shall address a different question, namely: Can the unit gamma model deliver a data fit that is superior to that obtained with the beta model? To that end, we shall use the data and regression model from Cribari-Neto and Souza [8]. The data are a cross section of 124 countries. Here, y is the proportion of atheists, x_2 is average IQ, x_3 is IQ squared, x_4 is a dummy variable that equals one if more than half of the population is Muslim, x_5 is per capita income (adjusted for purchasing power parity) and x_6 is the logarithm of the ratio between the sum of imports and exports and gross national product. The beta regression model used by the authors is

$$\begin{aligned}\log\log(\mu_i) &= \beta_1 + \beta_2 x_{i2} + \beta_3 x_{i3} + \beta_4 x_{i4} + \beta_5 x_{i5} + \beta_6 x_{i6} \\ \log(\phi_i) &= \gamma_1 + \gamma_2 x_{i2}.\end{aligned}$$

We estimated beta and unit gamma regressions with the above specification for μ_i and ϕ_i . At the outset, we note that since the two models share the same specification for both submodels (i.e. the same regressors and link functions), it is not possible to perform the J test as proposed by Cribari-Neto and Lucena [9,10]. It is possible, however, to perform other comparisons. For the beta (unit gamma) model, $\text{pseudo-}R_{FC}^2 = 0.638$ and $\text{pseudo-}R_N^2 = 0.760$ ($\text{pseudo-}R_{FC}^2 = 0.639$ and $\text{pseudo-}R_N^2 = 0.787$). We note that the $\text{pseudo-}R^2$ measures are slightly higher for the unit gamma model (3.6% larger when $\text{pseudo-}R_N^2$ is used). We now turn to the model selection criterion used in the previous application. For the beta (unit gamma) model, $\text{AIC} = -518.98$, $\text{AICc} = -517.73$ and $\text{BIC} = -496.42$ ($\text{AIC} = -548.24$, $\text{AICc} = -546.99$ and $\text{BIC} = -525.68$). The results clearly favor the unit gamma regression model. The differences in the corresponding criteria are nearly equal to 30 and we take that as strong evidence in favor of the unit gamma model. When the difference in the AIC values exceeds 4, one can conclude that there is considerably less support for the model with larger AIC; see Burnham and Anderson [7]. The difference in the AIC values here is over seven times larger than that threshold. Still according to Burnham and Anderson [7], when that difference exceeds 10, there is essentially no support for the model that displays the larger AIC value.

It is also interesting to analyze the z -tests p -values associated with the different regressors. For the test that $\mathcal{H}_0 : \beta_5 = 0$, we obtain the following p -values for the beta and unit gamma models: 0.0493 and 0.0149, respectively. It is well established in the literature that per capita income positively influences the prevalence of religious disbelievers; see, e.g.

[1,2]. That effect is only marginally statistically significant at the 5% significance level when the beta model is used, the evidence in favor of it being stronger in the unit gamma model. Even when the test of $\mathcal{H}_0 : \beta_5 = 0$ is carried out using the two corrected likelihood ratio test statistics, w^* and w^{**} , in both models, the evidence in favor of the statistical significance of the per capita income effect is stronger in the unit gamma model, where the tests p -values are over three times smaller (beta: 0.0643 and 0.0641, unit gamma: 0.0203 and 0.0202).

6. Concluding remarks

The unit gamma regression model can be used with responses that assume values in the standard unit interval, i.e. with dependent variables that assume values in $(0, 1)$, such as rates and proportions. It is an alternative to the beta regression model. The unit gamma regression model is based on the assumption that the variable of interest follows the unit gamma law which is parameterized in terms of mean and precision parameters. Inference on the parameters that index the model is typically carried out using the likelihood ratio test. Such a test, however, tends to be quite inaccurate in small samples. In particular, it tends to be liberal, i.e. it overrejects the null hypothesis when such a hypothesis is true. In this paper, we derived two modified likelihood ratio tests statistics that are expected to deliver more accurate inferences when the sample size is small.

We considered both fixed and varying precision unit gamma regression models. The latter contains a submodel for the response mean and a separate submodel for the precision whereas in the former only the dependent variable mean varies across observations. We derived two modified likelihood ratio test statistics (w^* and w^{**}) for the class of unit gamma regression models. We obtained results for both fixed and varying precision models. We performed Monte Carlo simulations to evaluate the tests finite sample behavior. The numerical evidence showed that the standard likelihood ratio testing inference can be quite inaccurate when the sample size is small. In particular, it can be considerably liberal. The two corrected tests, in contrast, tend to be much less size distorted, thus yielding more accurate inferences. The tests performances did not deteriorate when the mean responses assumed values close to one the standard unit interval limits. Our numerical results have also shown that one of the modified tests tends to be less liberal, i.e. more accurate: w^* .

We presented two empirical applications in which the variable of interest was modeled using both a beta regression model and a unit gamma regression model. In the first application, the corrected tests yielded an inference that was different from that obtained using the standard likelihood ratio test. Additionally, we found some evidence that the selected unit gamma regression model yields a slightly better data fit than the corresponding beta regression model. In the second application, the unit gamma model clearly outperformed the beta model.

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Appendix

In order to simplify the notation, we define the following matrices:

$$\begin{aligned}
 K_1 &= \text{diag} \left(\frac{d_i(1+d_i)}{\phi_i \mu_i^2} \left(\frac{1+2d_i}{\phi_i} - 1 \right) \right), \quad K_2 = \text{diag} \left(\frac{d_i(1+d_i)}{\mu_i^{2+1/\phi_i}} \right), \\
 K_3 &= \text{diag} \left(\frac{d_i}{\mu_i \phi_i} (1+d_i) \right), \\
 Z_1 &= \text{diag} \left(\frac{-d_i(1+d_i) \log(\mu_i)}{\mu_i \phi_i} \left(\frac{1+2d_i}{\phi_i^2} \right) - \frac{d_i(1+d_i)}{\mu_i \phi_i^2} \right), \\
 Z_2 &= \text{diag} \left(\frac{d_i(1+d_i)}{\mu_i \phi_i} \left(\frac{1}{d_i} + \frac{\log(\mu_i)}{\phi_i} + \frac{\log(\mu_i)}{\phi_i d_i} \right) \right), \\
 L_1 &= \text{diag} \left(\frac{d_i}{\phi_i^3} \left(\frac{\log(\mu_i)}{\phi_i} + \frac{d_i \log(\mu_i)}{\phi_i} + 2 \right) \log(\mu_i)(1+d_i) + \frac{d_i^2 (\log(\mu_i))^2}{\phi_i^4} (1+d_i) \right), \\
 L_2 &= \text{diag} \left(-\frac{d_i}{\phi_i^2} \log(\mu_i)(1+d_i) \left(\frac{1}{d_i} + \frac{\log(\mu_i)}{\phi_i} + \frac{\log(\mu_i)}{d_i \phi_i} \right) - \psi'(\phi_i) - \frac{\log(\mu_i)(1+d_i)}{\phi_i^2} \right),
 \end{aligned}$$

and

$$L_3 = \text{diag} \left(\frac{-d_i \log(\mu_i)(1+d_i)}{\phi_i^2} \right).$$

We shall now present details on the derivation of the correction used to define the improved test statistics. Our focus is, as noted earlier, on the unit gamma regression model. Similar derivations for the beta regression model can be found in Ferrari and Pinheiro [17].

Note that $\bar{q}$ is obtained from

$$q = \begin{bmatrix} \mathbb{E}[U_\beta(\theta_1)\ell(\theta_1)] - \mathbb{E}[U_\beta(\theta_1)\ell(\theta)] \\ \mathbb{E}[U_\delta(\theta_1)\ell(\theta_1)] - \mathbb{E}[U_\delta(\theta_1)\ell(\theta)] \end{bmatrix},$$

where $\theta = (\beta^\top, \delta^\top)^\top$.

Using Equations (4) and (7) we obtain

$$\begin{aligned} \mathbb{E}[U_\beta(\theta)\ell(\theta)] &= \mathbb{E}_\theta[X^\top T_1 \Phi^{-1} \mathcal{M}^{-1} D(\mathcal{I} + D)(y^* - \mu^*) \\ &\quad \times \{(y^* - \mu^*)^\top (D - \mathcal{I}) + (y^\dagger - \mu^\dagger)^\top (\Phi - \mathcal{I}) + b^\top\} \iota] \\ &= X^\top T_1 \Phi^{-1} \mathcal{M}^{-1} D(\mathcal{I} + D) \{\mathbb{E}_\theta[(y^* - \mu^*)(y^* - \mu^*)^\top] (D - \mathcal{I}) \\ &\quad + \mathbb{E}_\theta[(y^* - \mu^*)(y^\dagger - \mu^\dagger)^\top] (\Phi - \mathcal{I}) + \mathbb{E}_\theta[(y^* - \mu^*)] b^\top\} \iota \\ &= X^\top T_1 \Phi^{-1} \mathcal{M}^{-1} D(\mathcal{I} + D) \{V^*(D - \mathcal{I}) + C(\Phi - \mathcal{I})\} \iota. \end{aligned}$$

Since y_t and y_u , for $t \neq u$, are independent, and $\mathbb{E}_{\theta_1}(y_t^* - \mu_t^{*(1)}) = 0$, we have $\mathbb{E}_{\theta_1}[(y_t^* - \mu_t^{*(1)})(y_u^* - \mu_u^{*(1)})] = 0$. Also, $\mathbb{E}_{\theta_1}[(y_t^* - \mu_t^{*(1)})(y_t^* - \mu_t^*)] = \mathbb{E}_{\theta_1}[(y_t^* - \mu_t^{*(1)})(y_t^* - \mu_t^{*(1)})] + \mathbb{E}_{\theta_1}[(y_t^* - \mu_t^{*(1)})(\mu_t^{*(1)} - \mu_t^*)] = \mathbb{E}_{\theta_1}[(y_t^* - \mu_t^{*(1)})^2] = v_t^{*(1)}$. The superscript '(1)' indicates evaluation at θ_1 .

After some algebra, we obtain

$$\mathbb{E}_{\theta_1}[U_\beta(\theta_1)\ell(\theta)] = X^\top T_1^{(1)} \Phi^{-1(1)} \mathcal{M}^{-1(1)} D^{(1)} (\mathcal{I} + D^{(1)}) \{V^{*(1)}(D - \mathcal{I}) + C^{(1)}(\Phi - \mathcal{I})\} \iota.$$

Hence,

$$\begin{aligned} \mathbb{E}_{\theta_1}[U_\beta(\theta_1)\ell(\theta_1)] - \mathbb{E}_{\theta_1}[U_\beta(\theta_1)\ell(\theta)] &= X^\top T_1^{(1)} \Phi^{-1(1)} \mathcal{M}^{-1(1)} D^{(1)} (\mathcal{I} \\ &\quad + D^{(1)}) \{V^{*(1)}(D^{(1)} - D) + C^{(1)}(\Phi^{(1)} - \Phi)\} \iota. \end{aligned}$$

Similarly, using (4) and (8) we obtain

$$\begin{aligned} \mathbb{E}_\theta[U_\delta(\theta)\ell(\theta)] &= \mathbb{E}_\theta[H^\top T_2 \{P(y^* - \mu^*) + (y^\dagger - \mu^\dagger)\} \{(y^* - \mu^*)^\top (D - \mathcal{I}) + (y^\dagger - \mu^\dagger)^\top (\Phi - \mathcal{I}) + b^\top\} \iota] \\ &= H^\top T_2 \{P \mathbb{E}_\theta[(y^* - \mu^*)(y^* - \mu^*)^\top] (D - \mathcal{I}) + P \mathbb{E}_\theta[(y^* - \mu^*)(y^\dagger - \mu^\dagger)^\top] \times (\Phi - \mathcal{I}) \\ &\quad + \mathbb{E}_\theta[(y^\dagger - \mu^\dagger)(y^* - \mu^*)^\top] (D - \mathcal{I}) + \mathbb{E}_\theta[(y^\dagger - \mu^\dagger)(y^\dagger - \mu^\dagger)^\top] \times (\Phi - \mathcal{I})\} \iota \\ &= H^\top T_2 \{PV^*(D - \mathcal{I}) + PC(\Phi - \mathcal{I}) + C(D - \mathcal{I}) + V^\dagger(\Phi - \mathcal{I})\} \iota. \end{aligned}$$

Therefore,

$$\begin{aligned} \mathbb{E}_{\theta_1}[U_\delta(\theta_1)\ell(\theta_1)] - \mathbb{E}_{\theta_1}[U_\delta(\theta_1)\ell(\theta)] &= H^\top T_2^{(1)} \{(P^{(1)} V^{*(1)} + C^{(1)})(D^{(1)} - D) \\ &\quad + (P^{(1)} C^{(1)} + V^{\dagger(1)})(\Phi^{(1)} - \Phi)\} \iota, \end{aligned}$$

We shall now move to the derivation of $\tilde{\Upsilon}$ which is obtained from

$$\Upsilon = \begin{bmatrix} \mathbb{E}_{\theta_1}[U_\beta(\theta_1)U_\beta^\top(\theta)] & \mathbb{E}_{\theta_1}[U_\beta(\theta_1)U_\delta^\top(\theta)] \\ \mathbb{E}_{\theta_1}[U_\delta(\theta_1)U_\beta^\top(\theta)] & \mathbb{E}_{\theta_1}[U_\delta(\theta_1)U_\delta^\top(\theta)] \end{bmatrix}.$$

It follows from Equations (7) and (8) that

$$\begin{aligned}
\mathbb{E}_{\theta_1} [U_{\beta}(\theta_1) U_{\beta}^{\top}(\theta)] &= \mathbb{E}_{\theta_1} [X^{\top} T_1^{(1)} \Phi^{-1(1)} \mathcal{M}^{-1(1)} D^{(1)} (\mathcal{I} + D^{(1)}) (y^* - \mu^{*(1)}) \\
&\quad \times (y^* - \mu^*)^{\top} (\mathcal{I} + D) D \mathcal{M}^{-1} \Phi^{-1} T_1 X] \\
&= X^{\top} T_1^{(1)} \Phi^{-1(1)} \mathcal{M}^{-1(1)} D^{(1)} (\mathcal{I} + D^{(1)}) V^{*(1)} (\mathcal{I} \\
&\quad + D) D \mathcal{M}^{-1} \Phi^{-1} T_1 X, \\
\mathbb{E}_{\theta_1} [U_{\delta}(\theta_1) U_{\delta}^{\top}(\theta)] &= \mathbb{E}_{\theta_1} [H^{\top} T_2^{(1)} \{P^{(1)} (y^* - \mu^{*(1)}) + (y^{\dagger} - \mu^{\dagger(1)})\} \{(y^* \\
&\quad - \mu^*)^{\top} P^{\top} + (y^{\dagger} - \mu^{\dagger})^{\top}\} T_2 H] \\
&= H^{\top} T_2^{(1)} \{P^{(1)} V^{*(1)} P^{\top} + (P^{(1)} + P^{\top}) C^{(1)} + V^{\dagger(1)}\} T_2 H, \\
\mathbb{E}_{\theta_1} [U_{\beta}(\theta_1) U_{\delta}^{\top}(\theta)] &= \mathbb{E}_{\theta_1} [X^{\top} T_1^{(1)} \Phi^{-1(1)} \mathcal{M}^{-1(1)} D^{(1)} (\mathcal{I} + D^{(1)}) (y^* - \mu^{*(1)}) \\
&\quad \times \{(y^* - \mu^*)^{\top} P^{\top} + (y^{\dagger} - \mu^{\dagger})^{\top}\} T_2 H] \\
&= X^{\top} T_1^{(1)} \Phi^{-1(1)} \mathcal{M}^{-1(1)} D^{(1)} (\mathcal{I} + D^{(1)}) \{V^{*(1)} P^{\top} + C^{(1)}\} T_2 H, \\
\mathbb{E}_{\theta_1} [U_{\delta}(\theta_1) U_{\beta}^{\top}(\theta)] &= \mathbb{E}_{\theta_1} [H^{\top} T_2^{(1)} \{P^{(1)} (y^* - \mu^{*(1)}) + (y^{\dagger} - \mu^{\dagger(1)}) (y^* \\
&\quad - \mu^*)^{\top} (\mathcal{I} + D) \times D \mathcal{M}^{-1} \Phi^{-1} T_1 X] \\
&= H^{\top} T_2^{(1)} \{P^{(1)} V^{*(1)} + C^{(1)}\} (\mathcal{I} + D) D \mathcal{M}^{-1} \Phi^{-1} T_1 X.
\end{aligned}$$